LESSON 6 — Motor Power, Movement, and Telemetry

Predict, command, observe, and verify

MISSION Can you use code and telemetry to diagnose a mismatch one fact at a time?

Learning targets

CHECK	BY THE END OF THIS LESSON, I CAN...
[ ]	Identify object, method, and argument.
[ ]	Interpret sign and magnitude.
[ ]	Predict tank-drive motion.
[ ]	Compare commands, telemetry, and wheel behavior safely.

Words to know

VOCABULARY object | method | argument | setPower | magnitude | sign | command | telemetry | evidence

Code focus

```
leftMotor.setPower(leftPower);
rightMotor.setPower(rightPower);

telemetry.addData("Left Power", leftPower);
telemetry.addData("Right Power", rightPower);
telemetry.update();
```

Before the video: commit to a prediction

MY PREDICTION Predict the robot request for leftPower = +0.3 and rightPower = -0.3.

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Confidence before the lesson: 1 2 3 4 5 (circle one)

The POWERING ON safety contract

NON-NEGOTIABLE The sequence is completed before every live-motor test. If a step is skipped, remain disabled and restart at Step 1.

STEP	ACTION	TEAM CHECK
1	SECURE THE ROBOT Use stable blocks. Keep driven wheels and moving mechanisms clear of the table and floor.	[ ]
2	CONFIRM DISABLED Select the correct robot and OpMode, but do not press PLAY.	[ ]
3	LOOK AROUND Remove hands, tools, wires, scraps, and loose objects from moving areas.	[ ]
4	MAKE SURE ALL ARE CLEAR Everyone steps outside the boundary; secure hair, clothing, lanyards, strings, and jewelry.	[ ]
5	CALL AND WAIT Call POWERING ON! loudly. Pause long enough for everyone to stop, look up, and step away.	[ ]
6	VISUALLY CONFIRM, THEN ENABLE Scan the full area again. Press PLAY only when everyone and everything is clear.	[ ]

Callout script

BEFORE PLAY Robot is secured and DISABLED. Look around. Everyone clear? POWERING ON! ... Wait ... Area is visibly clear. Enabling now.

BEFORE CONTACT STOP pressed. Driver Station shows DISABLED. Hands may approach only after the operator confirms disabled.

Student agreement

I understand that anyone may call STOP and the operator will disable immediately. I will not approach or touch the robot until DISABLED is visually confirmed.

Student signature: ______________________________________ **Date:** _______________

Reference program: BasicTwoMotorTeleOp.java

Use this numbered reference throughout Lessons 1-6. The comments divide the program into startup and repeating control sections.

```
package org.firstinspires.ftc.teamcode;

import com.qualcomm.robotcore.eventloop.opmode.LinearOpMode;
import com.qualcomm.robotcore.eventloop.opmode.TeleOp;
import com.qualcomm.robotcore.hardware.DcMotor;

@TeleOp(name="Basic Two Motor TeleOp", group="Iterative Opmode")
public class BasicTwoMotorTeleOp extends LinearOpMode {
    private DcMotor leftMotor;
    private DcMotor rightMotor;

    @Override
    public void runOpMode() {
        leftMotor = hardwareMap.get(DcMotor.class, "left_motor");
        rightMotor = hardwareMap.get(DcMotor.class, "right_motor");

        // Set one motor REVERSE if the physical drivetrain requires it.
        // rightMotor.setDirection(DcMotor.Direction.REVERSE);

        telemetry.addData("Status", "Initialized");
        telemetry.update();
        waitForStart();

        if (opModeIsActive()) {
            while (opModeIsActive()) {
                double leftPower = -gamepad1.left_stick_y;
                double rightPower = -gamepad1.right_stick_y;
                leftMotor.setPower(leftPower);
                rightMotor.setPower(rightPower);
                telemetry.addData("Left Power", leftPower);
                telemetry.addData("Right Power", rightPower);
                telemetry.update();
            }
        }
    }
}
```

CODE DETECTIVE Circle waitForStart(). Box the while condition. Underline both setPower() calls. Draw a star beside each telemetry line.

LESSON 6 — Guided Video Notes

Motor Power, Movement, and Telemetry

DIRECTIONS At each timestamp, pause, think silently, discuss, and then record evidence from the code or whiteboard visual.

PAUSE	QUESTION	MY CODE / VISUAL EVIDENCE
0:28	Identify the object, method, and argument in setPower().	
0:38	What requests stop? What do sign and magnitude mean?	
0:48	What should equal positive side powers do?	
0:58	What should opposite signs do?	
1:08	What does telemetry help the team compare?	
1:47	What should happen when movement differs from prediction?	

MOST IMPORTANT IDEA The idea I would teach to a teammate is:

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LESSON 6	Hands-On Team Activity *Motor Power, Movement, and Telemetry*

Materials and setup

READY	ITEM
[ ]	Robot secured on stable blocks with both wheels clear.
[ ]	Prediction cards: 0.0/0.0, +0.3/+0.3, -0.3/-0.3, +0.3/-0.3, and +0.5/+0.2.
[ ]	Driver Station showing Left Power and Right Power telemetry.
[ ]	Printed predict-observe-verify sheet from this guide.
[ ]	POWERING ON safety poster visible at the test area.

Team roles

ROLE	JOB
Safety lead	Runs the POWERING ON and approach checkpoints.
Driver Station	Controls INIT, PLAY, and STOP.
Driver	Applies one gentle command at a time.
Left observer	Reports left-wheel direction and relative speed.
Right observer	Reports right-wheel direction and relative speed.
Telemetry reader	Reads commanded Left Power and Right Power.
Recorder	Captures prediction, command, observation, and result.
Debugger	Chooses the next single check after a mismatch.

Activity sequence

STEP	ACTION
1	Display leftMotor.setPower(leftPower); and ask students to identify the object, method, and argument.
2	Reveal one left/right pair at a time. Require a written prediction before any live command.
3	Run the full POWERING ON sequence, press PLAY, move the left stick gently, center it, then move the right stick gently.
4	Use gentle straight, backward, spin, and curve inputs. Pause at center between cases.
5	Press STOP and confirm DISABLED before the team approaches. Choose only one evidence-based check.

LESSON 6 — Lab Record and Exit Ticket

Motor Power, Movement, and Telemetry

Student/team: ______________________________ **Date:** _______________ **Robot:** ___________________

BEFORE PLAY	AFTER TEST
[ ] Robot secure [ ] DISABLED [ ] Area clear [ ] POWERING ON called [ ] Wait completed [ ] Visual clear	[ ] STOP pressed [ ] DISABLED visually confirmed Safety lead initials: __________

Predict - command - observe - verify

LEFT / RIGHT	PREDICTION	TELEMETRY + OBSERVATION	MATCH? / NEXT CHECK
0.0 / 0.0			
+0.3 / +0.3			
-0.3 / -0.3			
+0.3 / -0.3			
+0.5 / +0.2			
Student test: ____ / ____			

Debrief

#	QUESTION	MY RESPONSE
1	What does setPower(0.0) request?	
2	What do sign and magnitude represent?	
3	Why can telemetry and wheel motion disagree?	
4	What is the response to any mismatch?	

Exit ticket

ANSWER BEFORE LEAVING Telemetry shows +0.3 for a wheel that turns physically backward. List the first two facts you would verify after STOP and DISABLED.

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